Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 02)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

0 1 . 1 Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer ____ %

0 1 . 2 Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

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0 1 . 3 Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

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0 1 . 4 Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

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0 2 . 1 State the Brundtland Commission definition of sustainable development.

[1 mark]

0 2 . 2 Explain the difference between a negative feedback mechanism and a positive feedback mechanism, giving **one** example of each in the climate system.

[4 marks]

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0 2 . 3 Explain why a diverse ecosystem is more likely to resist change than an agroecosystem.

[1 mark]

0 2 . 4 Evaluate the extent to which copying the principles of natural systems could make human activities more sustainable.

[9 marks]

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0 3 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 3 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 3 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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0 4 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

0 4 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

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0 5 . 1 Explain why tropical rainforests have such high biodiversity.

[4 marks]

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0 5 . 2 Explain why rainforest species may be particularly vulnerable to environmental change.

[2 marks]

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0 5 . 3 Explain why rainforest plants and animals such as frogs are a rich source of medicines.

[2 marks]

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0 5 . 4 Explain the importance of tropical rainforest for the hydrological cycle and for soils.

[2 marks]

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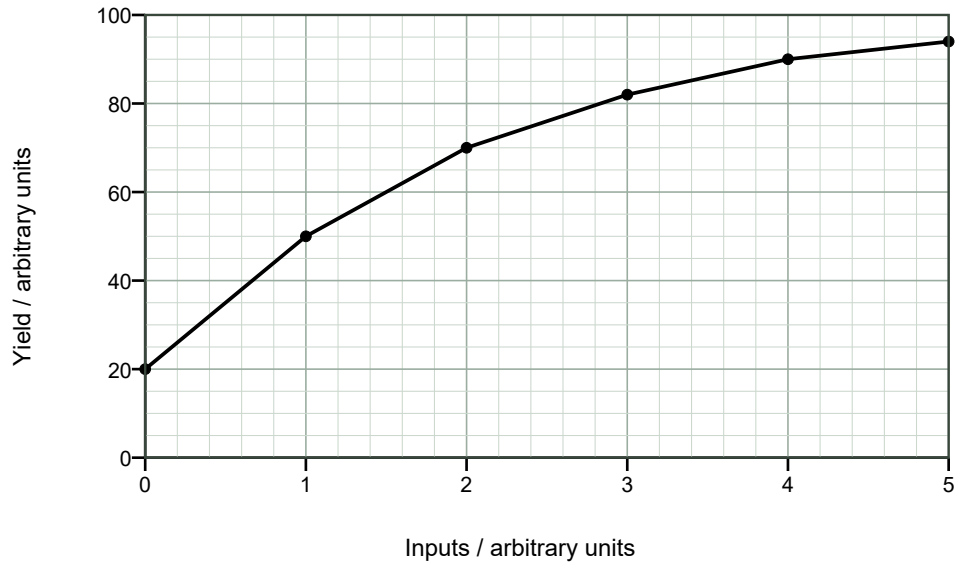
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0 6 Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1



0 6 . 1 Describe the relationship shown in Figure 1.

[2 marks]

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0 6 . 2 Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]

Answer _____ units

0 6 . 3 Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

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0 6 . 4 Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

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0 6 . 5 Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

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0 7 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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0 7 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 7 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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0 8 . 1 Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

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0 8 . 2 Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

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0 8 . 3 Describe how each of the following modern techniques provides information about past conditions.

[4 marks]

Isotope analysis of ice cores .....

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Gas analysis of ice cores .....

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Radiocarbon dating .....

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Magnesium : calcium ratio in calcite .....

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0 9 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 9 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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1 0 . 1 Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

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1 0 . 2 Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

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1 0 . 3 Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

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1 0 . 4 Evaluate the strategies that could be used to make agriculture more sustainable.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

OR

1 1 . 2 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS